Module-VII
Lecture-III
Fault Equivalence

1. Introduction

In the last lecture we learnt that structural testing with stuck-at fault model helps in reduction of the number of test patterns and also there is no requirement of DFT hardware. If there are n nets in a circuit then there can be $2n$ stuck-at faults and one test pattern can verify the presence/absence of the fault. So, the number of test patterns is liner in the number of nets in a circuit. Then we saw that one pattern can test multiple stuck-at faults, implying the total number of test patterns required is much lower than $2n$. In this lecture we will see if faults can be reduced, utilizing the fact that one pattern can test multiple faults and retaining any of these faults would suffice. Let us consider the example of an AND gate in Figure 1, with all possible stuck-at-0 faults. To test the fault at I1, input pattern is I1=1,I2=1; if the output is 0, s-a-0 fault in I1 is present, else it is absent. Now, also for the s-a-0 fault in net I2, the pattern is I1=1,I2=1. Same, pattern will test the s-a-0 fault in the output net O. So, it may be stated that although there are three s-a- 0 faults only one pattern can test them. In other words, keeping one fault among these three would suffice and these faults are equivalent.

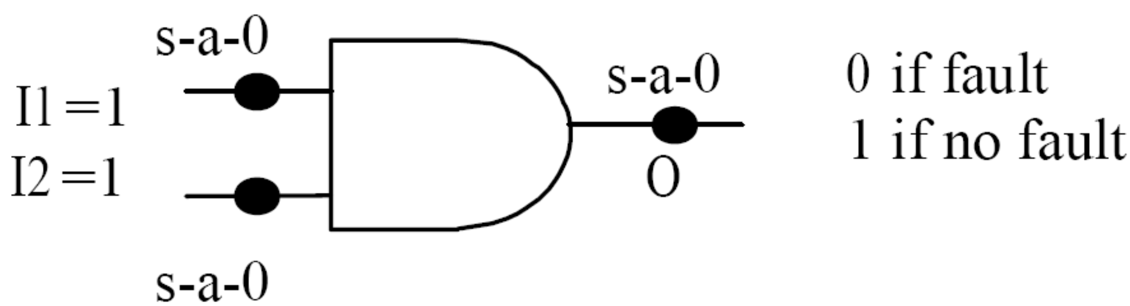


Figure 1. Stuck-at-0 faults in an AND gate

In the next section, we will see in detail how equivalent faults can be found in a circuit and then collapse them to reduce the number of patterns required to test them.

2. Fault Equivalence For Single Stuck-At Fault Model

Two stuck-at faults *f1* and *f2* are called equivalent iff the output function represented by the circuit with *f1* is same as the output function represented by the circuit with *f2*. Obviously, equivalent faults have exactly the same set of test patterns. In the AND gate of Figure 1, all stuck-at-0 faults are equivalent as they transform the circuit to O=0 and have I1=1, I2=1 as the test pattern.

Now lets us see what stuck-at faults are equivalent in other logic gates. Figure 2 illustrates equivalent faults in AND, NAND, OR, NOR gates and Inverter with the corresponding test patterns. The figure is self explanatory and can be explained from the discussion of AND gate (Figure 1)

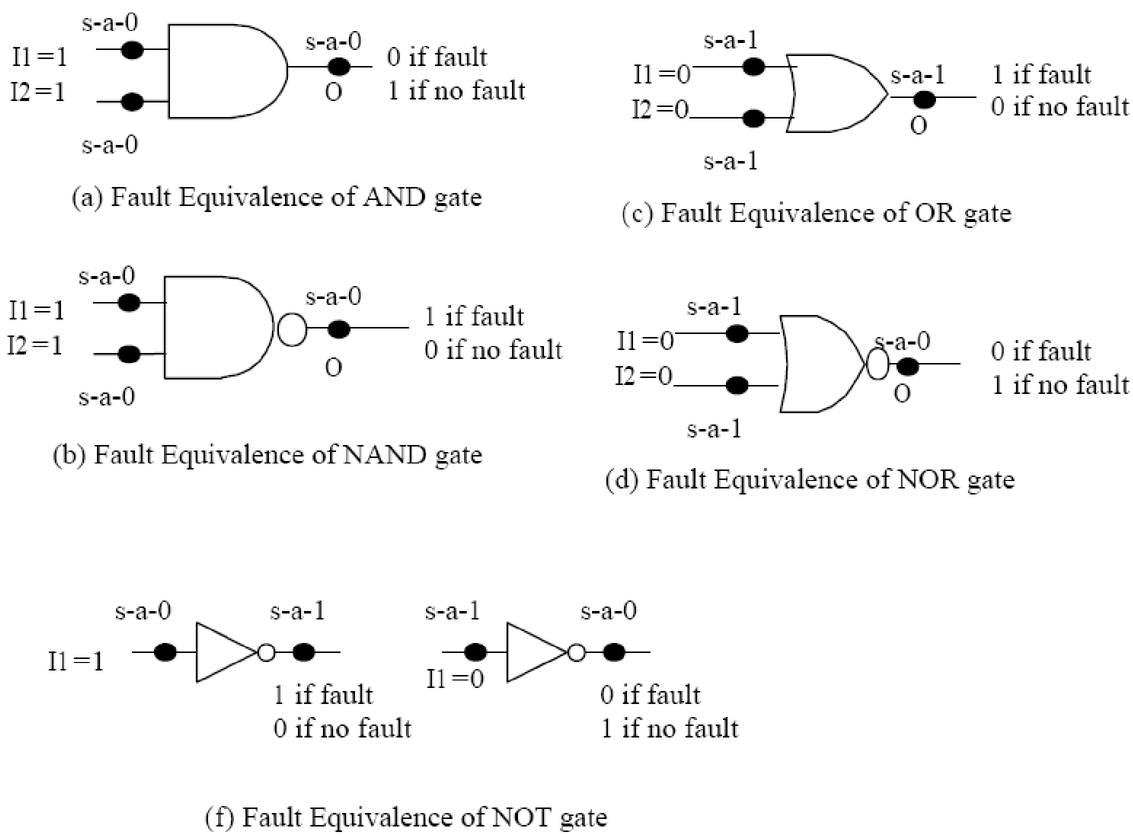


Figure 2. Equivalent faults in AND, NAND, OR, NOR gates and inverter

However, an interesting case occurs for fanout nets. The faults in the stem and branches are not equivalent. This is explained as follows. Figure 3 shows the stuck-at faults in a fanout which drives two nets and the corresponding test patterns.

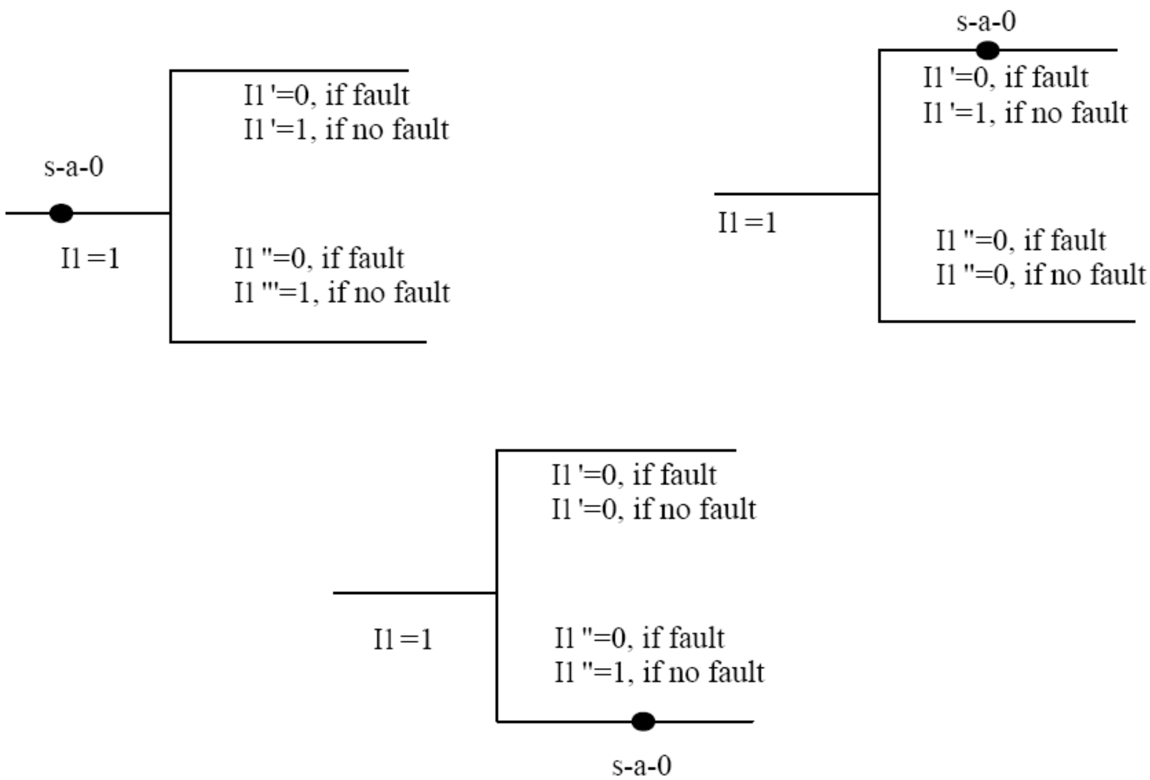
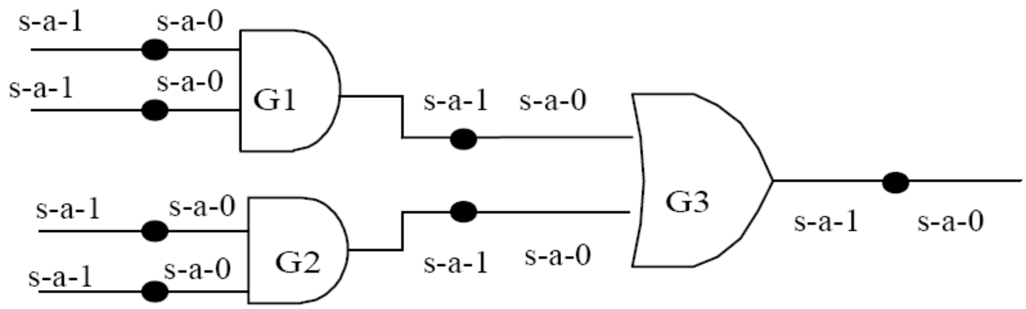


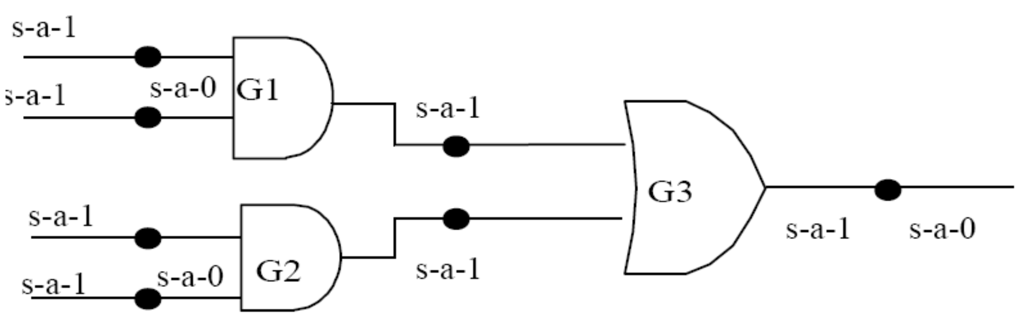
Figure 3. No equivalence of stuck-at fault in fanout net

It may be noted that s-a-0 fault in the stem and the two branches have the same test pattern (I1=1), however these three faults do not result in same change of the output functions of the stem and the braches. For example, if s-a-0 fault is in the stem, function of stem is I1=0 and for the branch I1’ (I1”) function is I1’=0 (I1”=0). So, fault at stem results in same output functions of the stem and the branches. However, if fault is in branch I1’, then function of stem is I1=0/1 (as driven) and for the branch I1’ (I1”) function is I1’=0 (I1”=I1). So output function of I1’ is different from I1 and I1”. Similar logic would hold for s-a-1 fault.

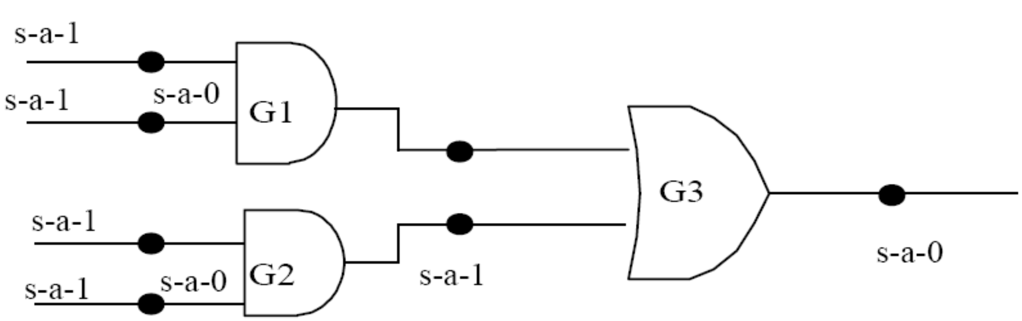
Now we consider a circuit and see the reduction in the number of faults by collapsing using fault equivalence. Figure 4 gives an example of a circuit without any fanout and illustrates the step wise collapsing of faults. In a circuit, collapsing is done level wise as discussed below.



(a) Circuit with all possible stuck at faults



(b) Circuit after collapsing equivalent faults at level 1



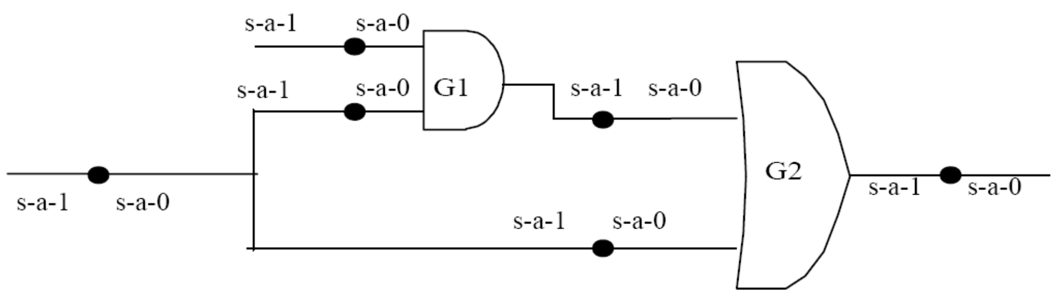
(c) Circuit after collapsing equivalent faults at level 2

Figure 4. Circuit without fanout and step wise collapsing of equivalent faults.

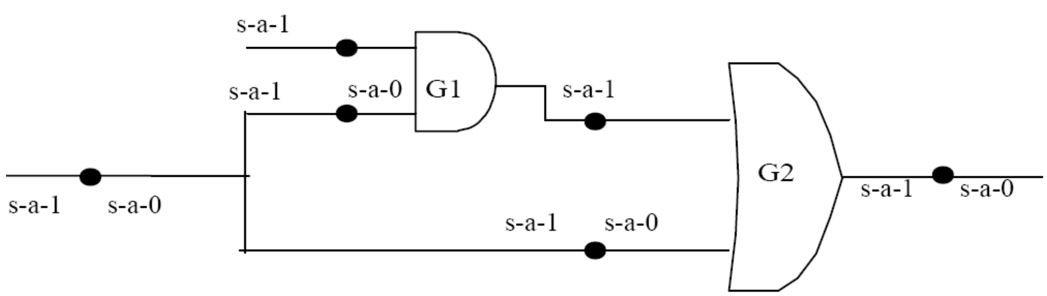
- Step1: Collapse all faults at level-1 gates (G1 and G2). In the AND gate G1 (and G2) the s-a-0 faults at output and one input (first) are collapsed; s-a-0 fault is retained only at one input (second).
- Step-2: Collapse all faults at level-2 gates (G3). In the OR gate G3 the s-a-1 faults at output and one input (from G1) are collapsed; s-a-1 fault is retained only at one input (from G2).

Now we consider a circuit with fanout. Figure 5 gives an example of a circuit with a fanout driving two nets and illustrates the step wise collapsing of faults.

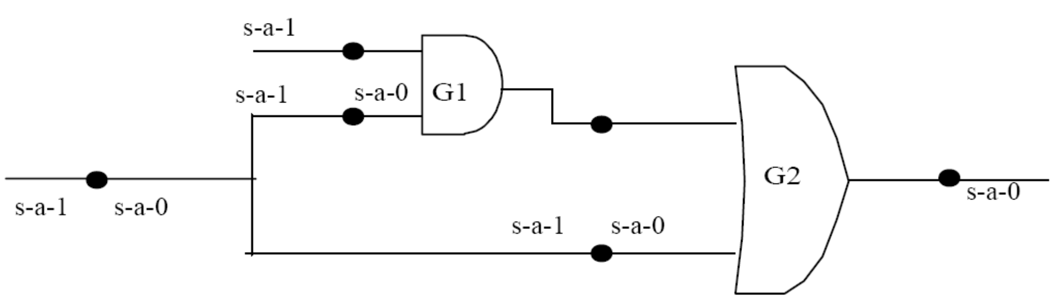
- Step1: Collapse all faults at level-1 gate (G1). In the AND gate G1 the s-a-0 faults at output and one input (first) are collapsed; s-a-0 fault is retained only at one input (second).
- Step-2: Collapse all faults at level-2 gates (G2). In the OR gate G2 the s-a-1 faults at output and one input (from G1) are collapsed; s-a-1 fault is retained only at one input (from fanout net).



(a) Circuit (having fanout) with all possible stuck at faults



(b) Circuit after collapsing equivalent faults at level 1



(c) Circuit after collapsing equivalent faults at level 2

Figure 5 Circuit with fanout and step wise collapsing of equivalent faults.

Is collapsing by fault equivalence the only way to reduce the number of faults? The answer is no. In the next section we will discuss another paradigm to reduce stuck-at faults—by fault dominance

4. Fault Dominance For Single Stuck-At Fault Model

If all tests of a stuck-at fault *f1* detect fault *f2* then *f2* dominates *f1*. If *f2* dominants *f1* then *f2* can be removed and only *f1*is retained.

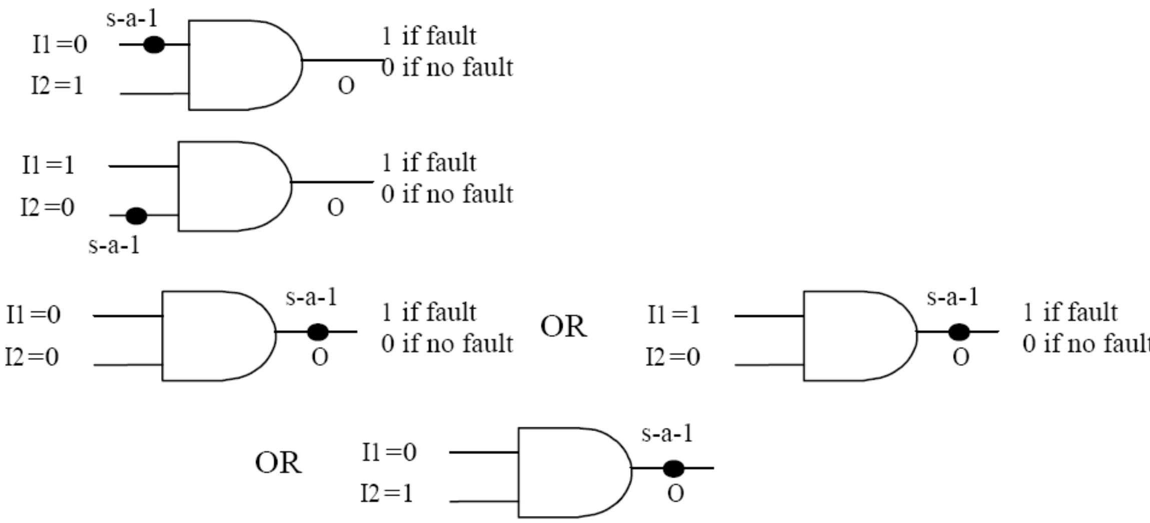
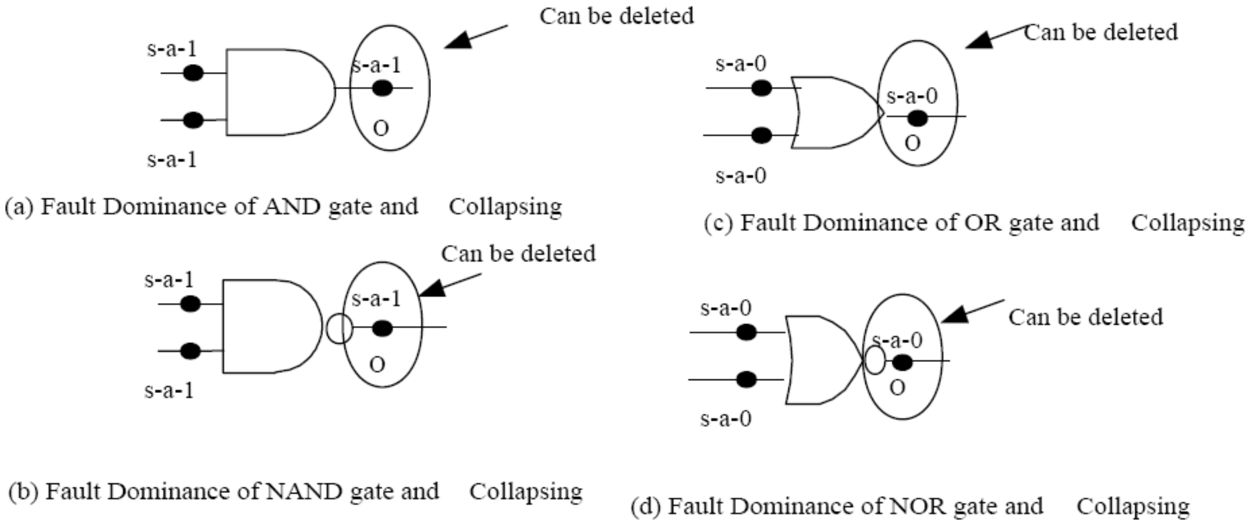


Figure 6. Fault collapsing for AND gate using dominance

Let us consider the example of an AND gate with s-a-1 faults in the inputs and output. To test the s-a-1 fault in the input I1 of the gate, test pattern is I1=0,I2=1. Similarly, for the s-a-1 fault in the input I2 of the gate, test pattern is I1=1,I2=0. But for the s-a-1 fault in the output of the gate, test pattern can be one of the three (i) I1=0,I2=0, (ii) I1=1,I2=0 (iii) I1=0,I2=1. So s-a-1 fault at the output dominates the s-a-1 faults at the inputs. So collapsing using dominance would result in dropping the s-a-1 fault at the output and retaining only the faults at in inputs.

The basic logic behind collapsing using dominance can be explained in simple logic as follows: If one retains the s-a-1 fault at the output and uses pattern I1=0,I2=0 to test it, then s-a-1 faults in the inputs are left untested. So faults at the inputs cannot be collapsed. However, if s-a-1 faults at the inputs are retained, then patterns 1=1,I2=0 and I1=0,I2=1 are applied, thereby testing the s-a-1 at the output twice. So, s-a-1 fault at the output can be collapsed and retain the ones at the inputs.

Figure 7, illustrates dominant faults in AND, NAND, OR, and NOR gates and fault which can be collapsed.



(a) Fault Dominance of AND gate and Collapsing

(c) Fault Dominance of OR gate and Collapsing

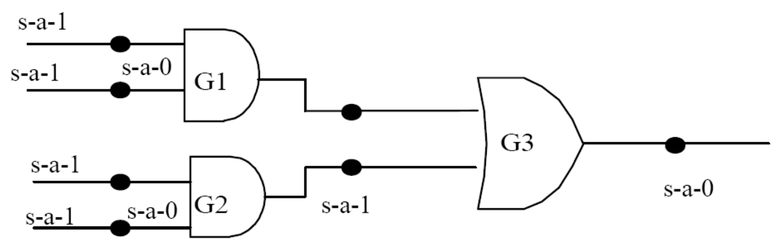
(b) Fault Dominance of NAND gate and Collapsing

(d) Fault Dominance of NOR gate and Collapsing

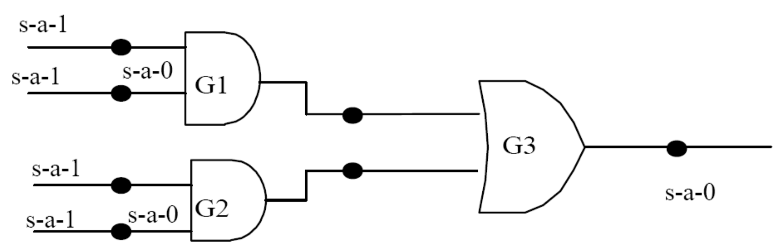
Figure 7. Dominant faults in AND, NAND, OR, and NOR gates

Now we consider the circuits illustrated in Figure 4 and Figure 5 and see further reduction in the number of faults by collapsing using fault dominance. Figure 8 illustrates fault collapsing using dominance for the circuit without fanout. As in case of equivalence, collapsing using dominance is done in level wise as discussed below.

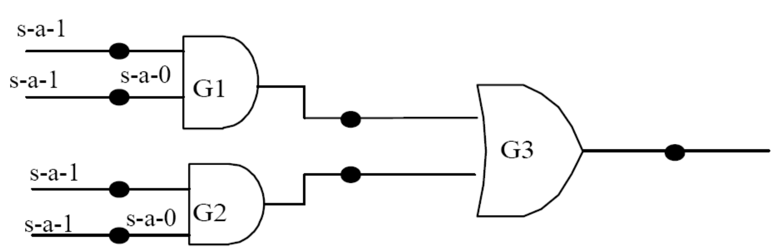
- Step1: Collapse all faults at level-1 gates (G1 and G2). In the AND gate G1 (and G2) the s-a-1 faults at output is collapsed; s-a-1 faults is retailed only at the inputs.
- Step-2: Collapse all faults at level-2 gates (G3). In the OR gate G3 the s-a-0 fault at output is collapsed and the s-a-0 faults at the inputs are retained. *It may be noted that the s-a-0 faults at inputs of G3 are retained indirectly by s-a-0 faults at the inputs of G1 and G2 (using fault equivalence).*



(a) Circuit after collapsing all equivalent faults



(b) Circuit after collapsing all dominating faults at level 1

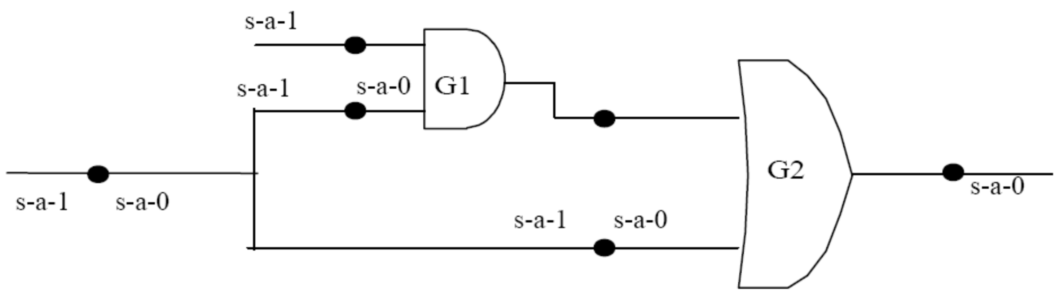


(c) Circuit after collapsing all dominating faults at level 2

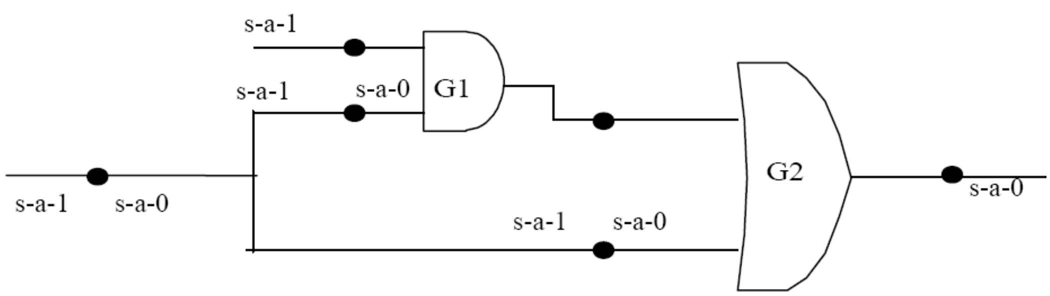
Figure 8. Circuit without fanout and step wise collapsing of faults using dominance.

Now we consider the circuit with fanout given in Figure 5 and see reduction in the number of faults by collapsing using fault dominance (in Figure 9). The steps are as follows

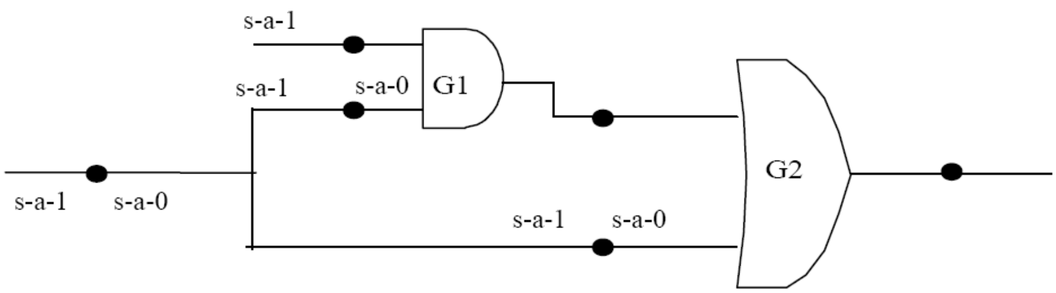
- Step1: Collapse all faults at level-1 gate (G1). No faults can be collapsed.
- Step-2: Collapse all faults at level-2 gates (G2). In the OR gate G2 the s-a-0 fault at the output is collapsed, as s-a-0 fault at the inputs are retained; s-a-0 fault from the fanout branch is retained explicitly and the one at the output of gate G1 is retained indirectly by s-a-0 faults at the inputs of G1 (using fault equivalence).



(a) Circuit (having fanout) after collapsing equivalent faults



(b) Circuit (having fanout) after collapsing all dominated faults at level 1



(c) Circuit (having fanout) after collapsing all dominated faults at level 2

Figure 9. Circuit with fanout and step wise collapsing of faults using dominance.

The following can be observed after faults are collapsed using equivalence and dominance:

1. A circuit with no fanouts, s-a-0 and s-a-1 faults is to be considered only at the primary inputs (Figure 8(c)). So in a fanout free circuit test patters are 2. (Number of primary inputs).
2. For circuit with fanout, checkpoints are primary inputs and fanout branches. Faults are to be kept only on the checkpoints (Figure 9(c)). So a test pattern set that detects all single stuck-at faults of the checkpoints detects all single stuck-at faults in that circuit.

Points 1 and 2 are termed as check point theorem.

Question and Answers:

1. For what class of circuits, maximum benefit is achieved due to fault collapsing and when the benefits are less? What is the typical number for test patterns required to test these classes of circuits?

Answer1.

For circuit with no fanouts, maximum benefit is obtained because faults in the primary inputs cover all other internal faults. So total number of test vectors are 2.(number of primary inputs).

In a circuit with a lot of fanout branches, minimum benefit is obtained as faults need to be considered in all primary inputs and fanout branches. So total number of test vectors are 2.(number of primary inputs + number of fanout branches).

2. What faults can be collapsed by equivalence in case of XOR gate?

Answer 2.

From the figure given below it may be noted that all stuck-at faults in the inputs and output of a 2-input XOR gate, results in different output function. So fault collapsing cannot be done for XOR gate using fault equivalence.